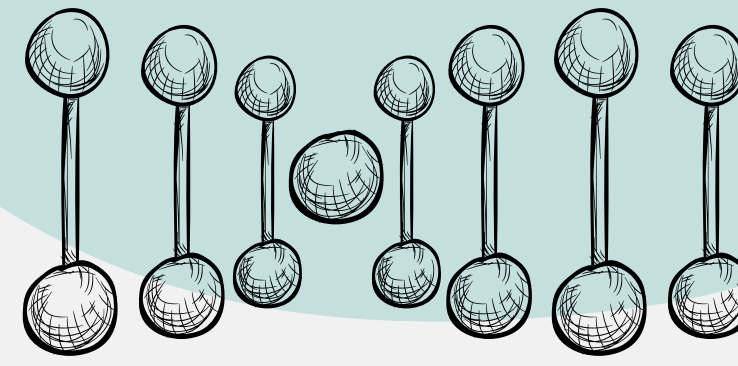
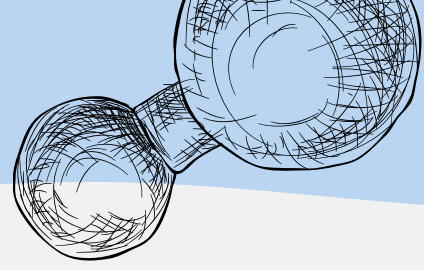
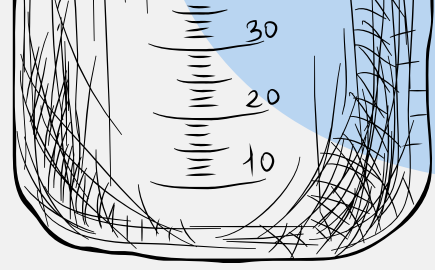
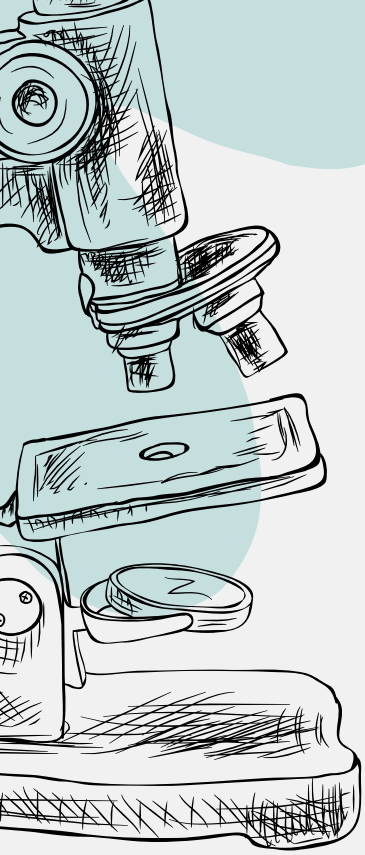
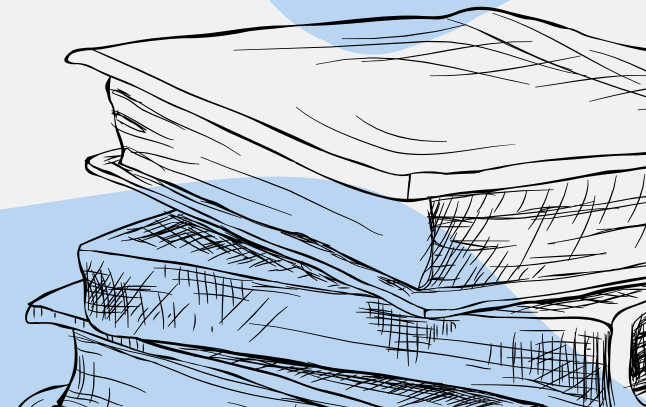
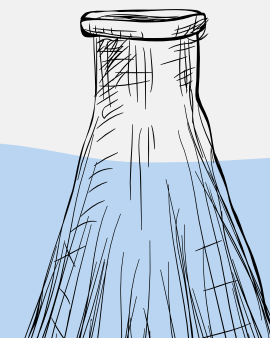
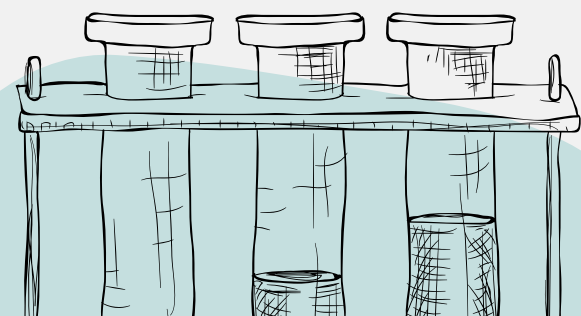
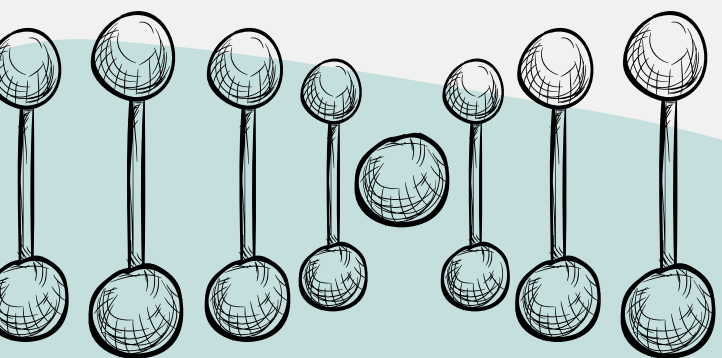
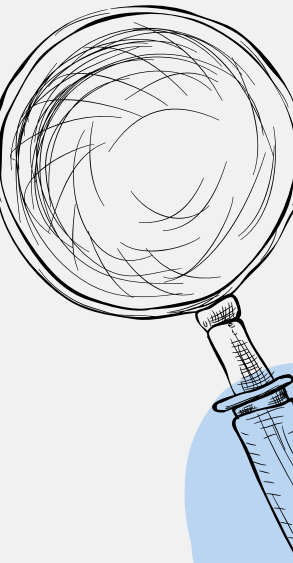




GenIO: Leveraging LLM Advancements in the Detection, Analysis, and Filling of Gaps During DNA Sequencing

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Project Overview

01

Introduction
and Motivation

02

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Gap-Filling Pipeline

06

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01

Introduction and Motivation

Problem:

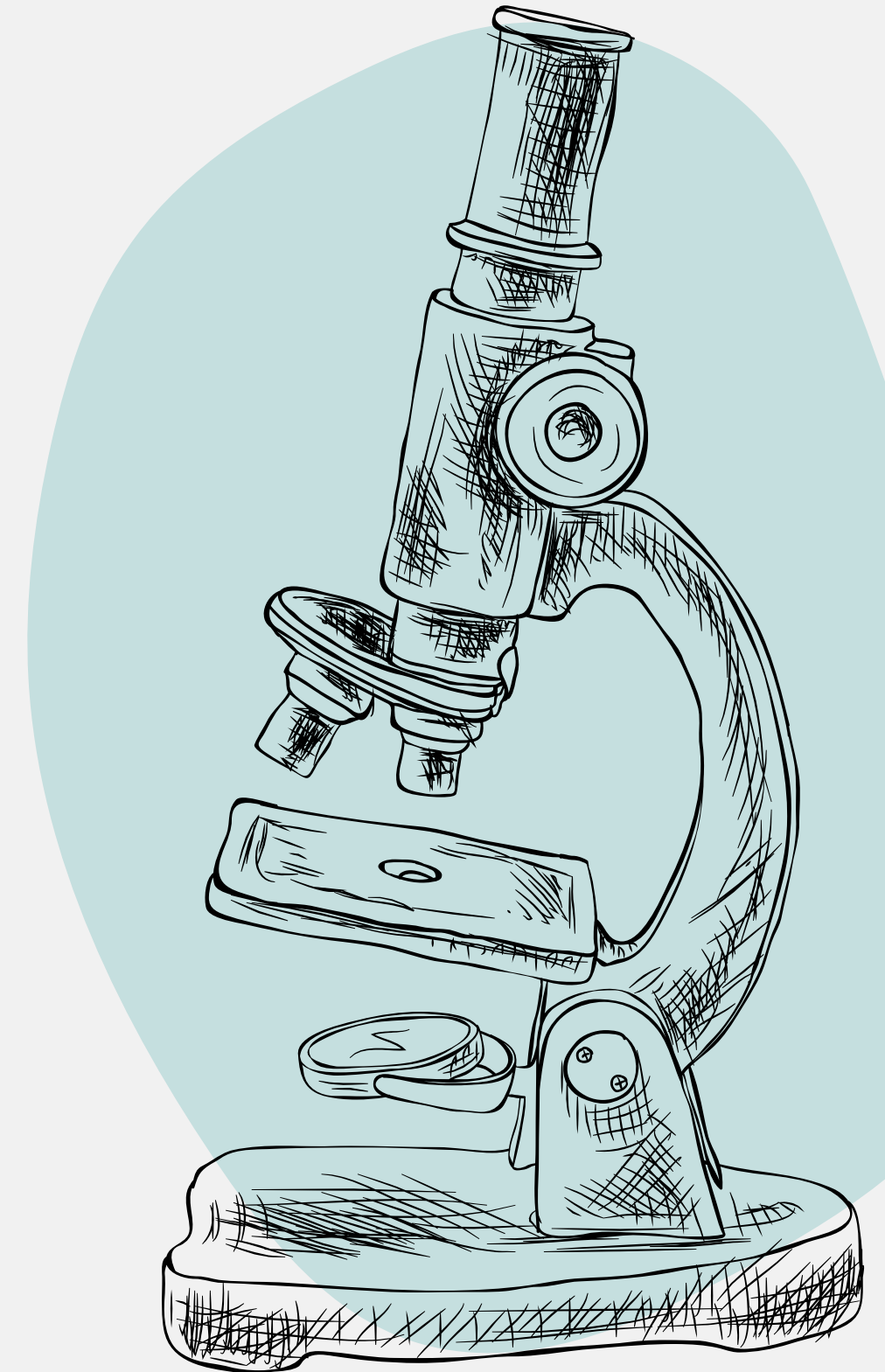
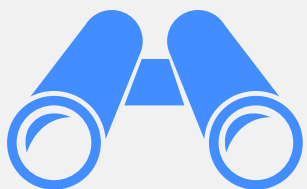
- Multi-step process
- Gaps due to sequencing limitations
- Manual or algorithmic gap-filling can be error-prone
- FASTA are not optimized

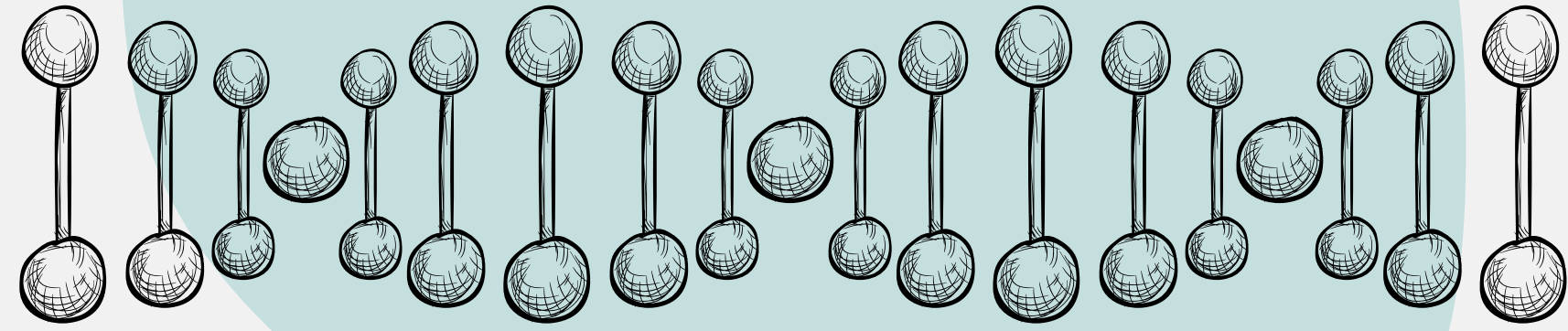
Motivation:

- LLMs have shown success in tasks requiring sequence prediction and contextual understanding
- Recent breakthroughs in bioinformatics highlight their potential in genome analysis

Objective:

- Improve genomic reconstruction in a more scalable, automated, and accurate way





02 Genomic LLM Overview

01

Architecture

02

Tokenization

03

Context-Length

04

Pre-Training

05

Species

Descriptor	Release Date	DNA Source	Technology	Language LLM	Fine-tuned/ Pretrained	Tokenization	Context Length
BigBird [36]	Jan-21	Human	Transformer	Yes	Yes	BPE	36 Kbp
gapseq [40]	Mar-21	Bacteria	Linear programming	No	No	No	Entire genome
GapPredict [5]	Sep-21	Human	LSTM	No	Yes	one-hot encoding	500 bp
GeneBERT [22]	Oct-21	Human	Encoder-only	Yes	Yes	k-mer	1 Kbp
Enformer [1]	Oct-21	Multispecies	Transformer	Yes	Yes	one-hot encoding	200 Kbp
EpigenomicBERT [33]	Dec-21	Human	Encoder-only	Yes	Yes	k-mer	1000 bp
LookingGlass [15]	May-22	Bacteria/Archaea	LSTM	No	Yes	Nucleotide-level	136 bp
Figbird [29]	Jun-22	Human/Bacteria	Probabilistic EM	No	No	No	Dynamic
ViBE [14]	Jun-22	Virus	Encoder-only	Yes	Yes	k-mer	510 bp
LOGO [35]	Aug-22	Human	Encoder-only	Yes	Yes	k-mer	2 Kbp
FloraBERT [17]	Aug-22	Plant	Encoder-only	Yes	Yes	BPE	1 Kbp
INHERIT [2]	Aug-22	Bacteria/Virus	Encoder-only	Yes	Yes	k-mer	500 bp
GenSLMs [41]	Nov-22	Virus/Bacteria	Decoder-only	Yes	Yes	k-mer	Entire genome
Geneformer [31]	May-23	Human	Encoder-only	Yes	Yes	custom tokenizer	200 Kbp
GPN [4]	Apr-23	Multispecies	CNN	Yes	Yes	Nucleotide-level	512 bp
DNAGPT [37]	Jul-23	Multispecies	Decoder-only	Yes	Yes	k-mer	13.5 Kbp
HyenaDNA [24]	Nov-23	Human	Decoder-only	Yes	Yes	Nucleotide-level	1 Mbp
megaDNA [28]	Oct-23	Bacteria	Decoder-only	Yes	Yes	Nucleotide-level	96 Kbp
GPN-MSA [3]	Oct-23	Multispecies	Transformer	Yes	Yes	Nucleotide-level	128 bp
UTR-LM [8]	Oct-23	Multispecies	Transformer	Yes	Yes	Nucleotide-level	128 bp
DLGapCloser [7]	Aug-24	Fungi/Plant	Bi-LSTM	No	Yes	one-hot encoding	2000 bp
DNABERT-2 [39]	Mar-24	Multispecies	Encoder-only	Yes	Yes	BPE	10 kbp
SegmentNT [10]	Mar-24	Human	Encoder-only	Yes	Yes	k-mer	50 Kb
Caduceus [27]	Jun-24	Human	BiMamba + RC	Yes	Yes	Nucleotide-level	131 Kb
AgroNT [21]	Jul-24	Plant	Encoder-only	Yes	Yes	k-mer	6 Kb
GROVER [26]	Jul-24	Human	Encoder-only	Yes	Yes	BPE	2.1 Kb
DNAGRINDER [38]	Sep-24	Multispecies	Encoder-only	Yes	Yes	BPE	12 Kbp
GenomicLLM [19]	Feb-24	Human	Transformer	Yes	Yes	SentencePiece	300 bp
EpiGePT [12]	Dec-24	Multispecies	Encoder-only	Yes	Yes	one-hot encoding	128 Kbp
SpliceBERT [6]	May-24	Multispecies	Encoder-only	Yes	Yes	Nucleotide-level	1024 bp
Evo [23]	Nov-24	Bacteria/Archaea	Transformer	Yes	Yes	Nucleotide-level	131 Kb
LLaMA-Gene [18]	Nov-24	Multispecies	Decoder-only	Yes	Yes	BPE	300-1000 bp
GENA-LM [11]	Jan-25	Human	Encoder-only	Yes	Yes	BPE	36 Kbp
BarcodeMAE [25]	Feb-25	Insects	Transformer	Yes	Yes	k-mer	768 tokens
Nucleotide Transformer (NT) [9]	Feb-25	Multispecies	Encoder-only	Yes	Yes	k-mer	12 Kb

Model Selection and Benchmarking Setup

Model	Date	AR	MLM	Tokenization	Context	Target	Architecture
HyenaDNA	Nov-23	✓	✗	Nucleotide	Long	Human	Decoder
megaDNA	Oct-23	✓	✗	Nucleotide	Long	Bacteria	Decoder
DNABERT-2	Mar-24	✗	✓	BPE	Long	Multispecies	Encoder
GENA-LM	Jan-25	✗	✓	BPE	Long	Human	Encoder-only
DNAGRINDER	Sep-24	✗	✓	BPE	Long	Multispecies	Encoder
BarcodeMAE	Feb-25	✗	✓	k-mer	Short	Insects	Transformer
GapPredict	Sep-21	✓	✗	One-hot	Short	Human	LSTM
DLGapCloser	Aug-24	✓	✗	One-hot	Short	Fungi/Plant	Bi-LSTM
Figbird	Jun-22	N/A	N/A	No	Short	Human/Bacteria	Probabilistic
NT	Feb-25	✗	✓	k-mer	Long	Multispecies	Encoder
AgroNT	Jul-24	✗	✓	k-mer	Long	Plant	Encoder
SegmentNT	Mar-24	✗	✓	k-mer	Long	Human	Encoder
GROVER	Jul-24	✗	✓	BPE	Long	Human	Encoder
Evo	Nov-24	✓	✗	Nucleotide	Long	Bacteria/Archaea	Transformer
Caduceus	Jun-24	✗	✓	Nucleotide	Long	Human	BiMamba + RC

04

Evaluation of Models' Performance and Analysis of Results (1)

Dataset:

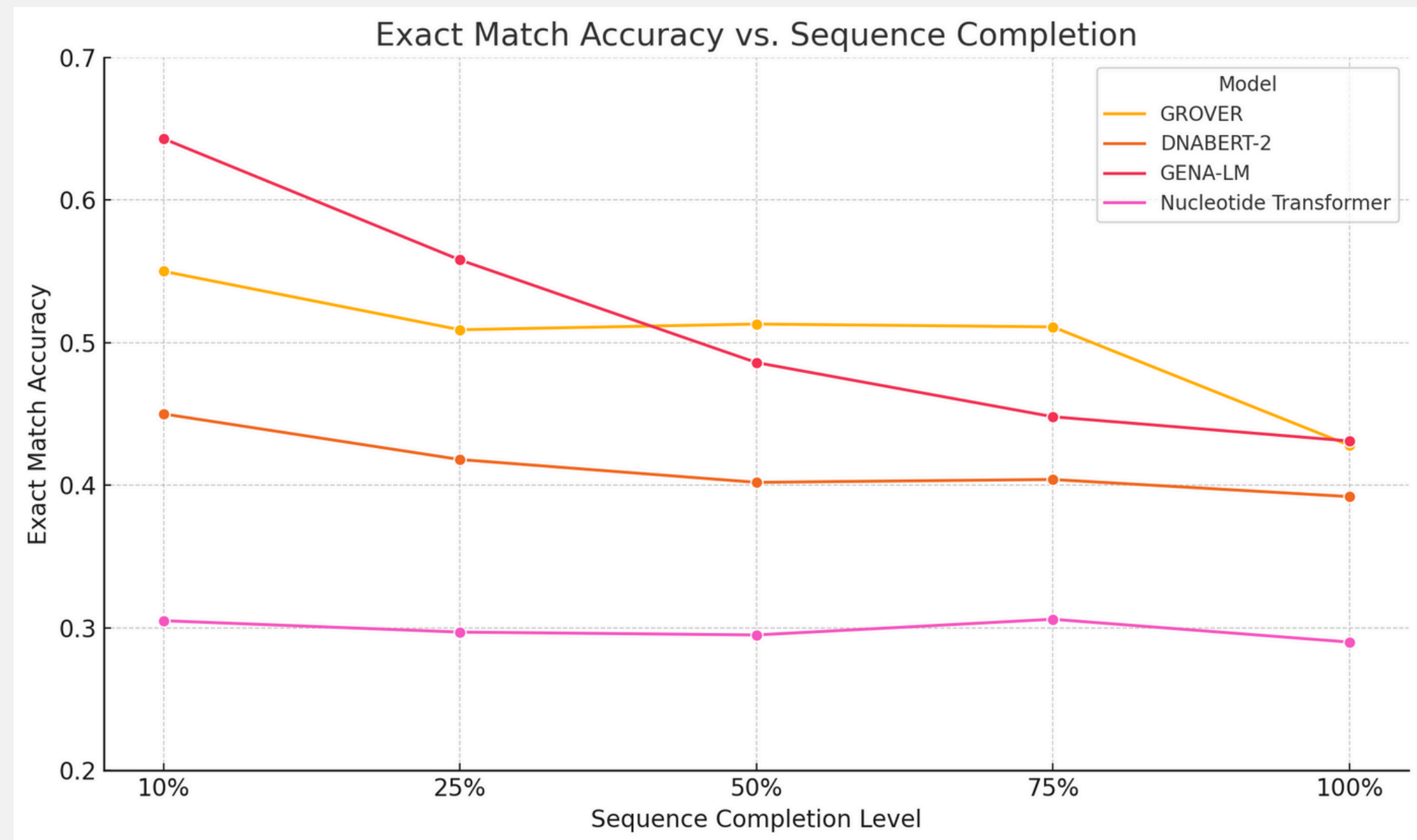
- 381 genomes from all 73 bacterial phyla available in the NCBI database
- Divided in CONTEXT and TARGET
- Sequence lengths: 1000bp, 2000bp and 5000bp

Environment:

- Local CPU setup (11th Gen Intel Core i7-1165G7 @ 2.80GHz, 16 GB RAM)

Methodology:

- Token-by-token prediction
- Top-1 Accuracy and Top-60 Accuracy



Results for 1000bp Context Length

04

Evaluation of Models' Performance and Analysis of Results (2)

Environment:

- Google Colab GPU T4

Methodology:

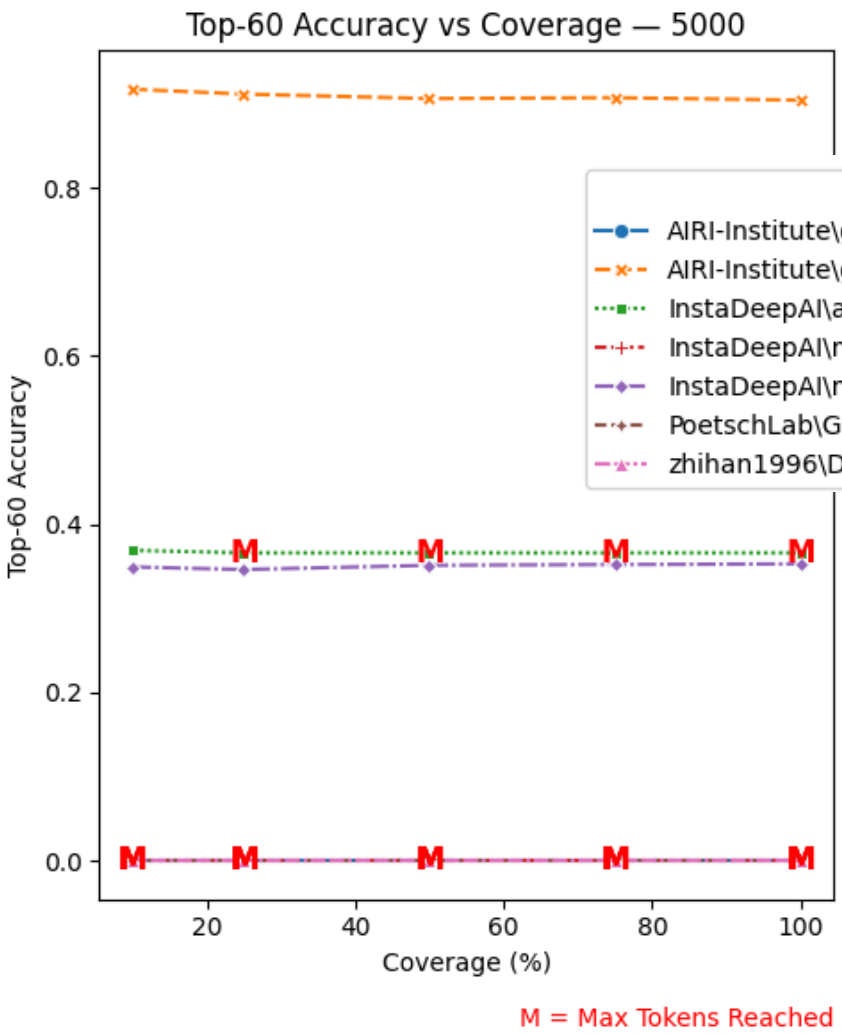
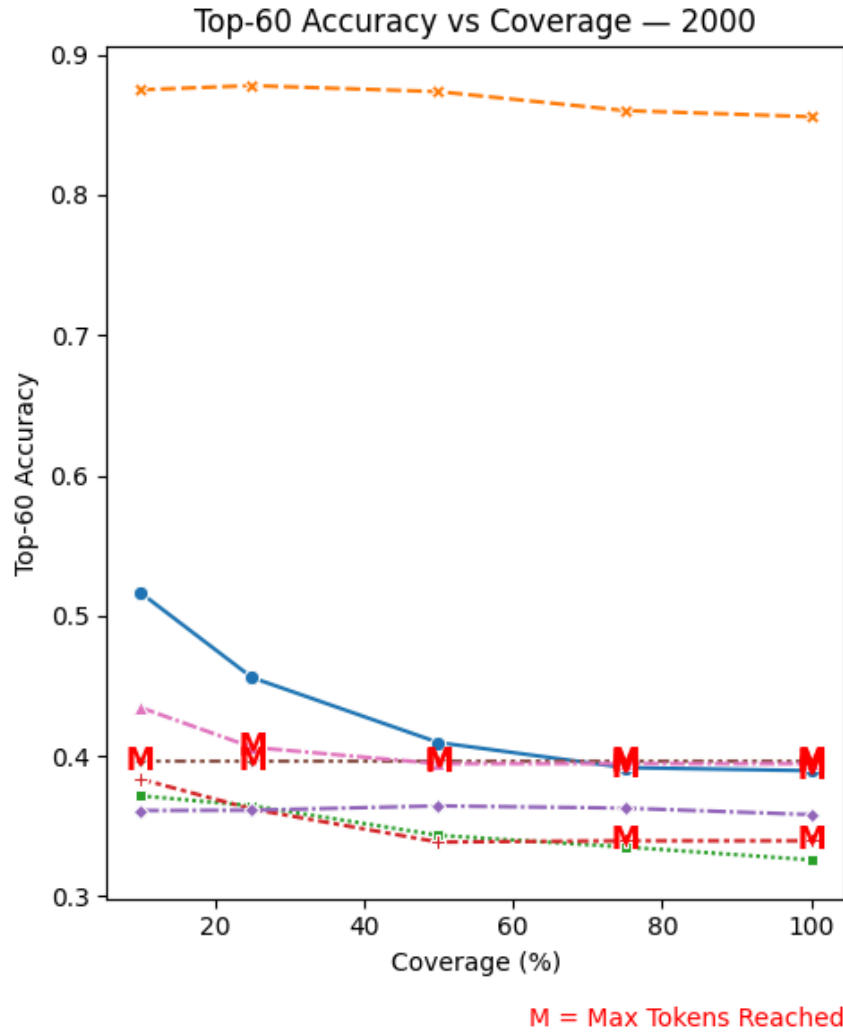
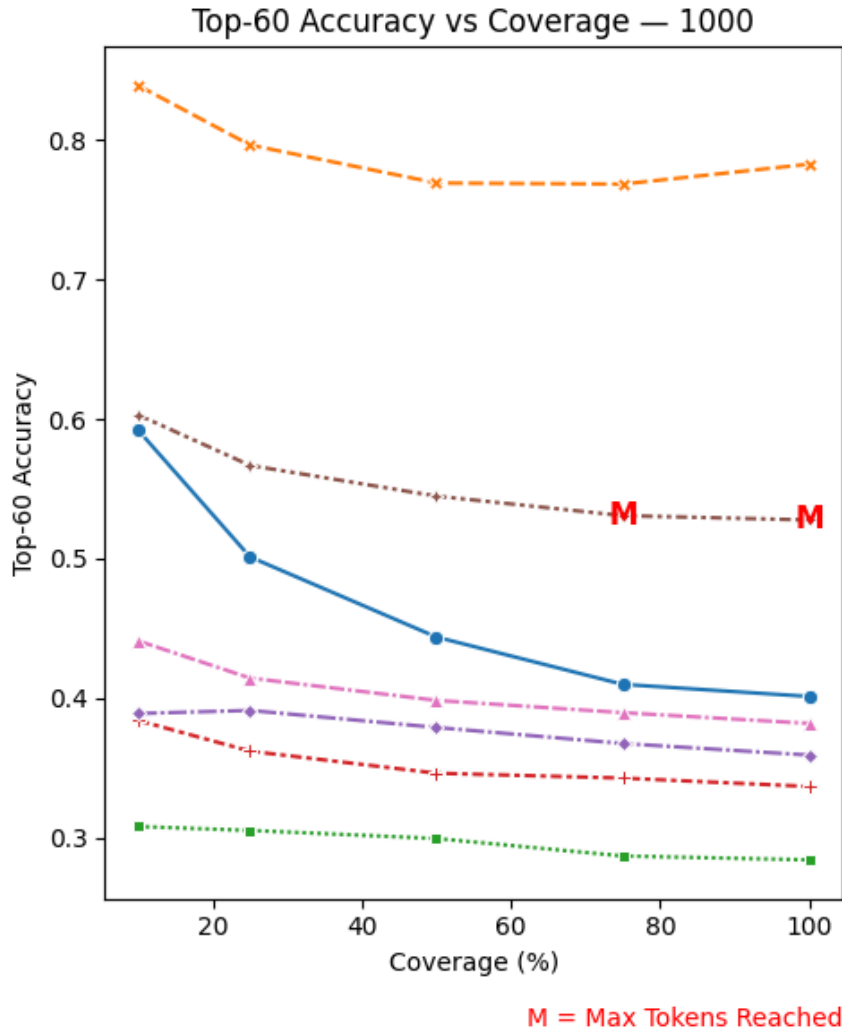
- Token-by-token prediction
- Standard script for all models

Metrics:

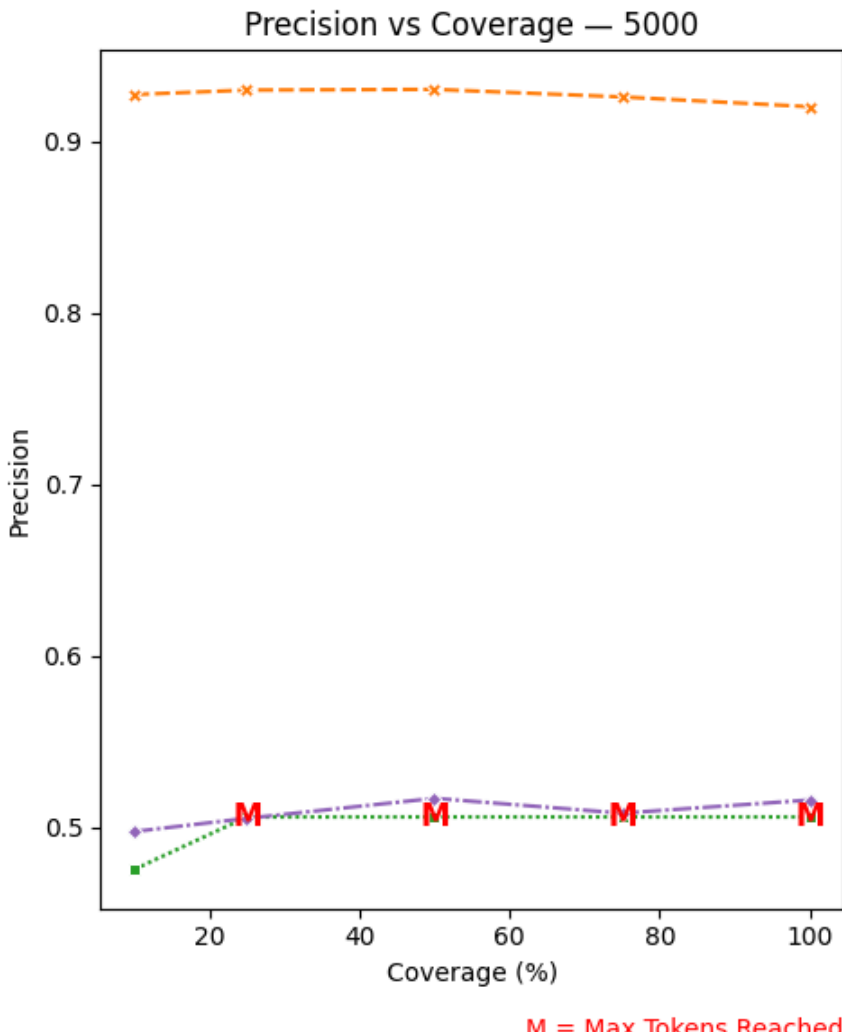
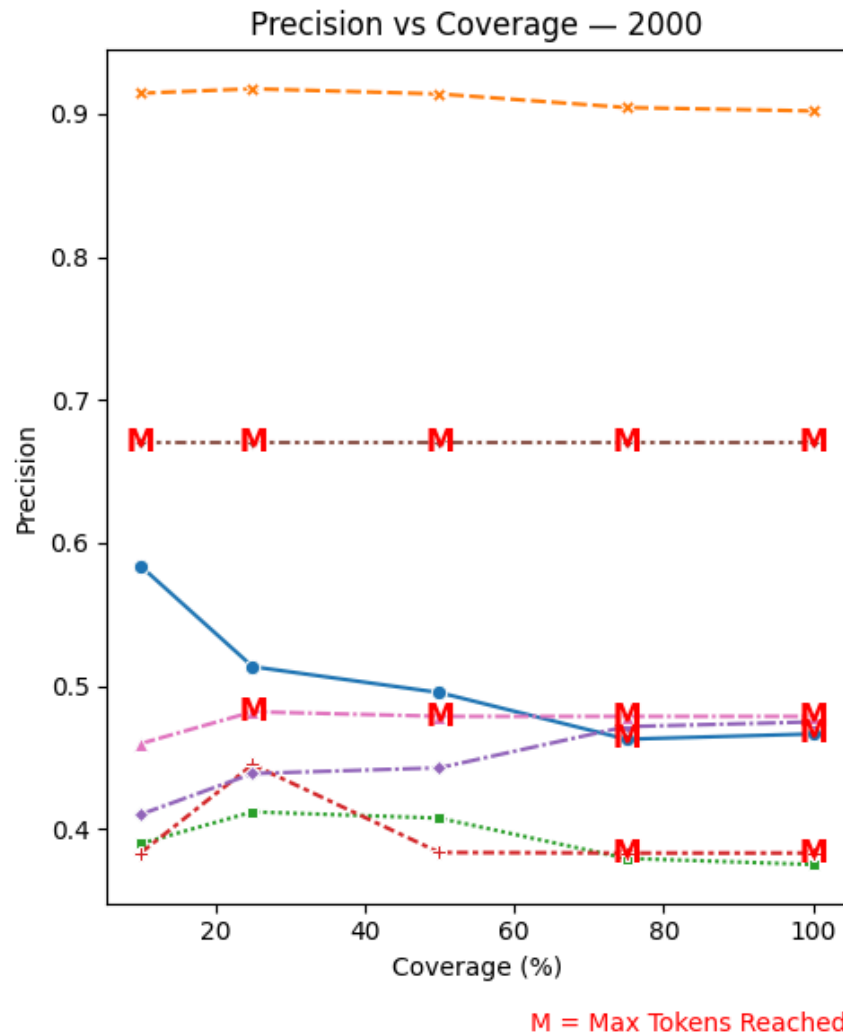
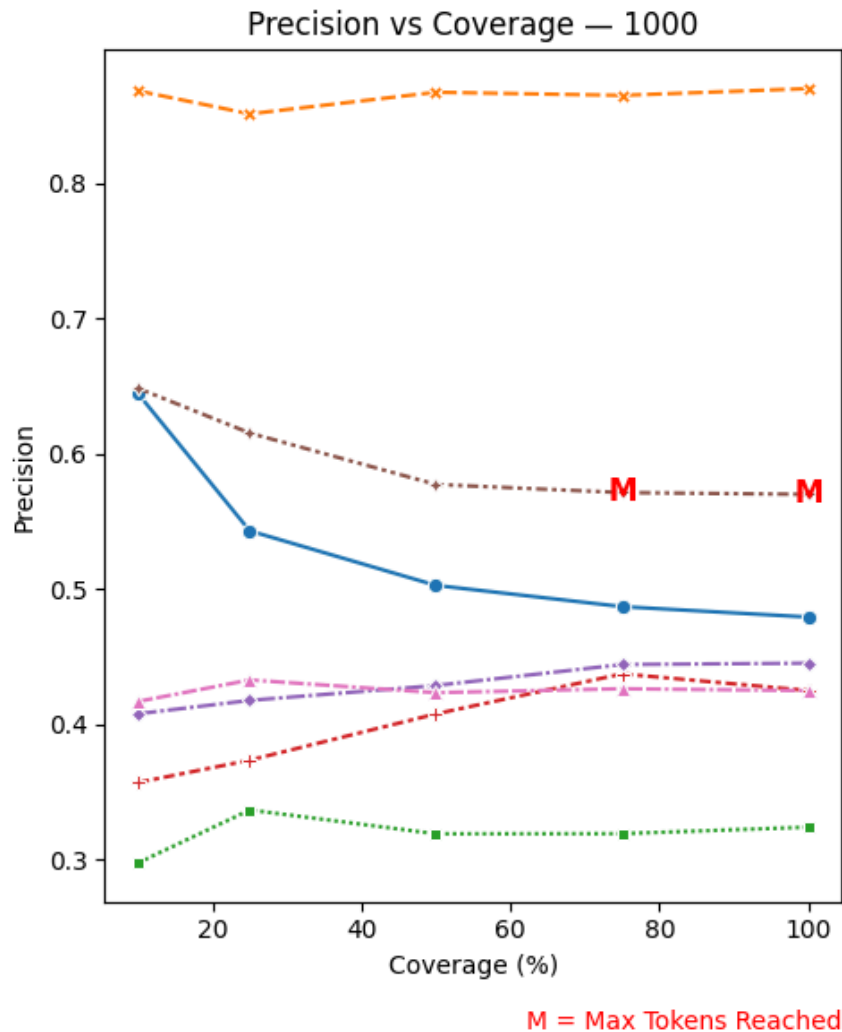
- Top-1 Accuracy and Top-60 Accuracy
- Precision, Recall, and F1 Score
- Levenshtein Distance
- Computational metrics



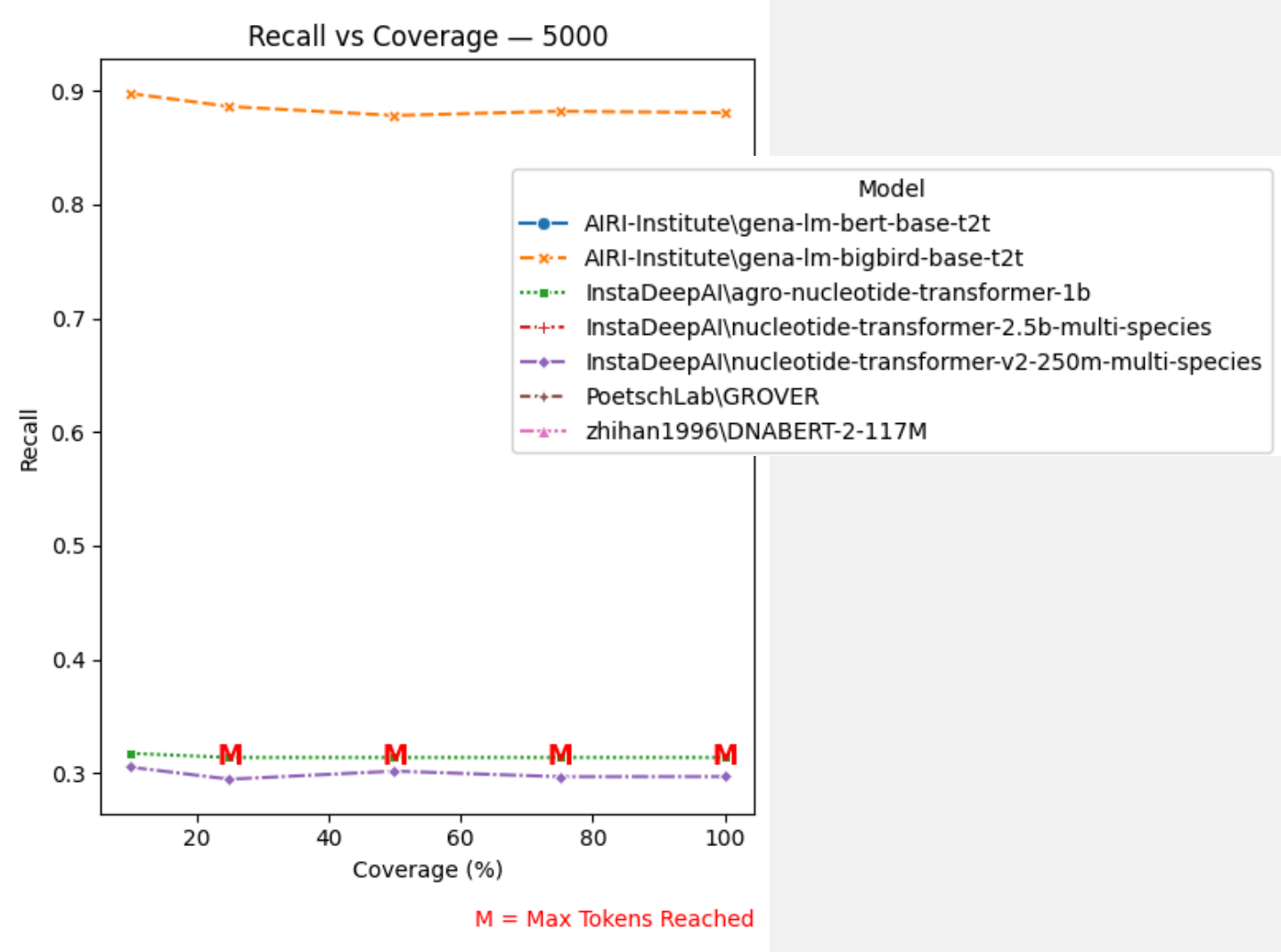
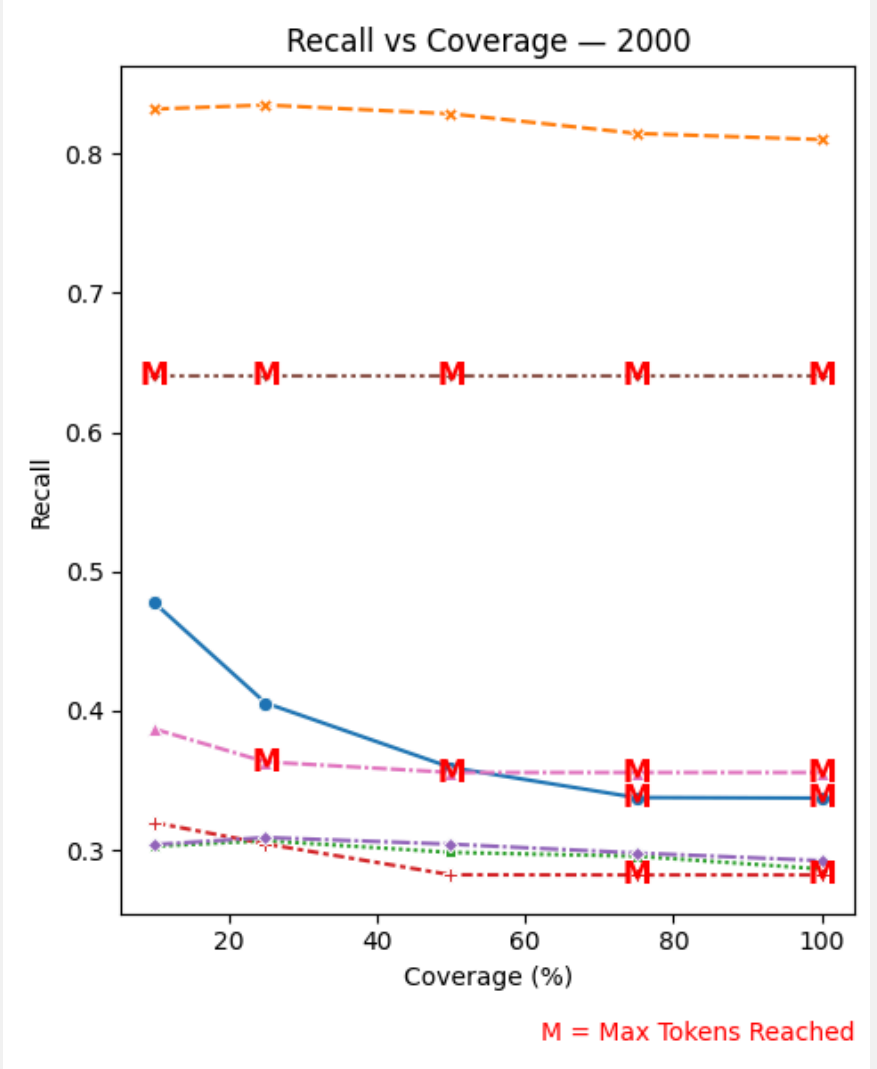
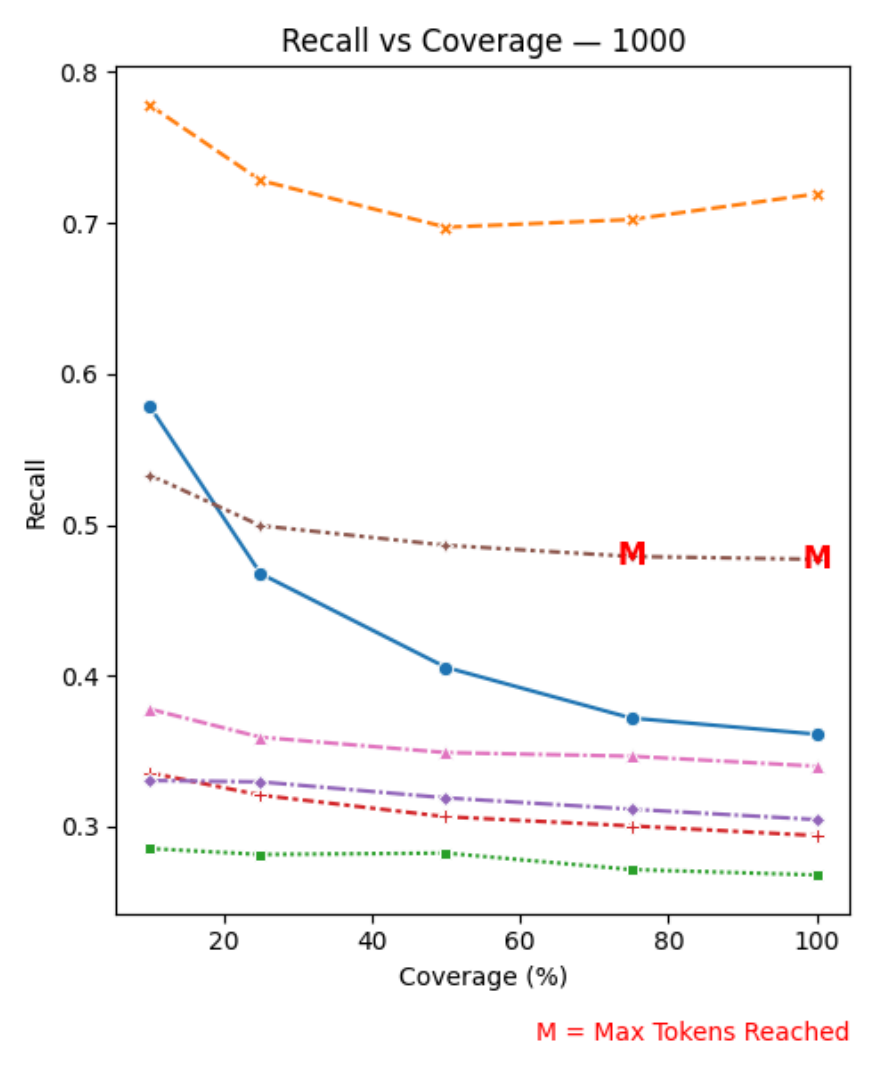
Top-60 Accuracy



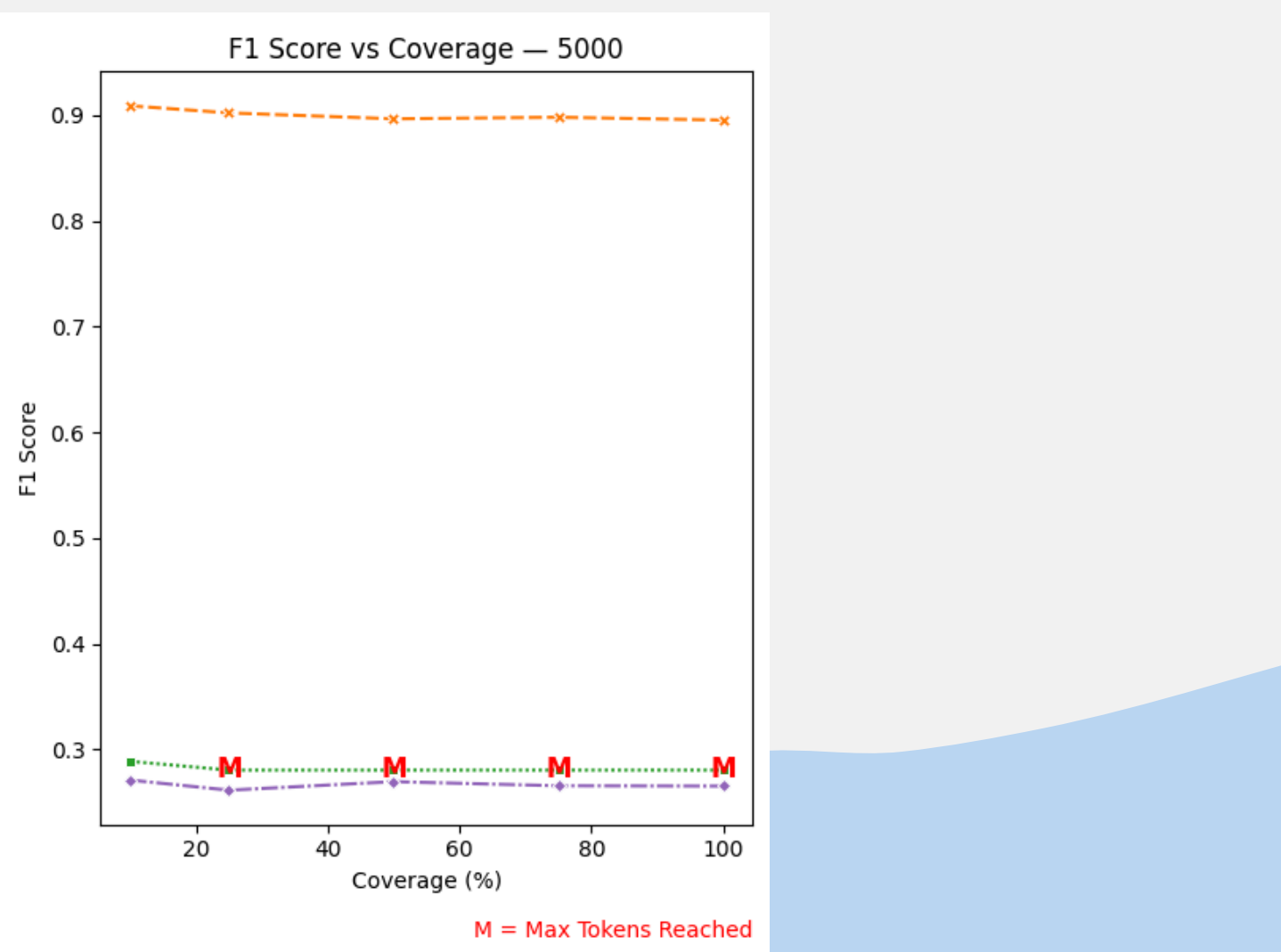
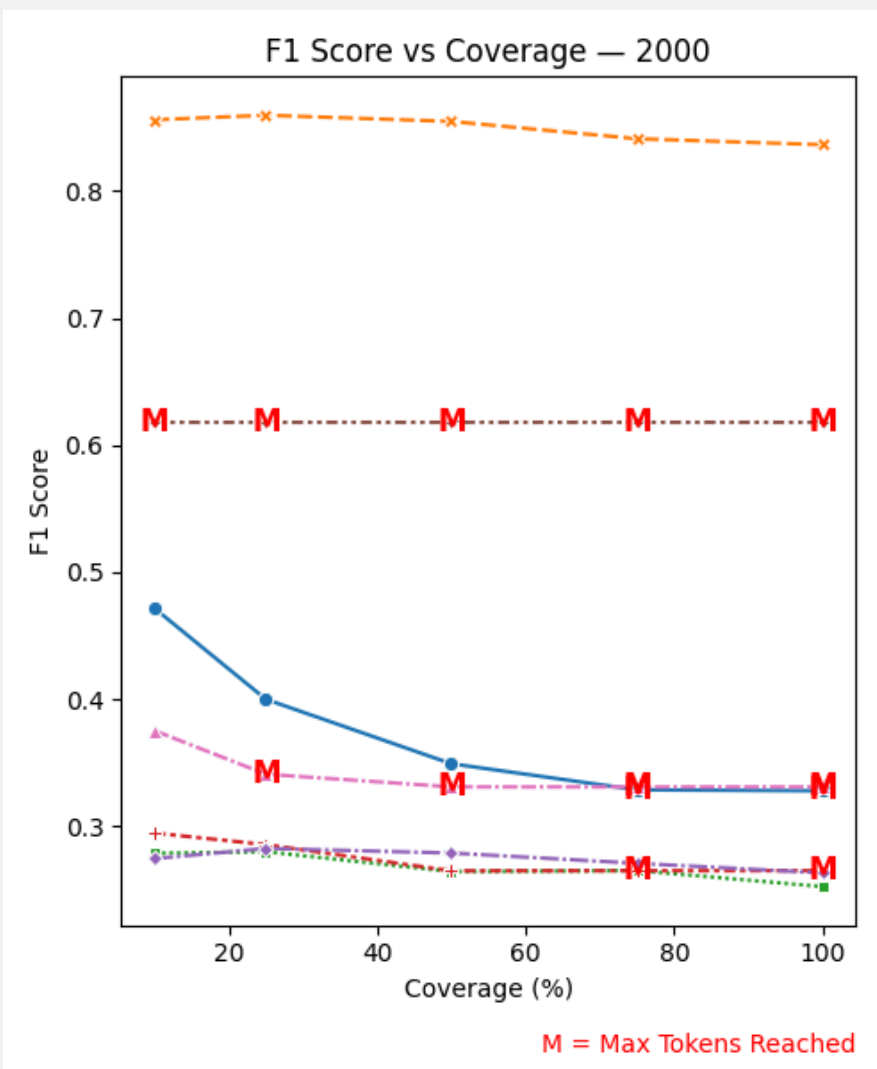
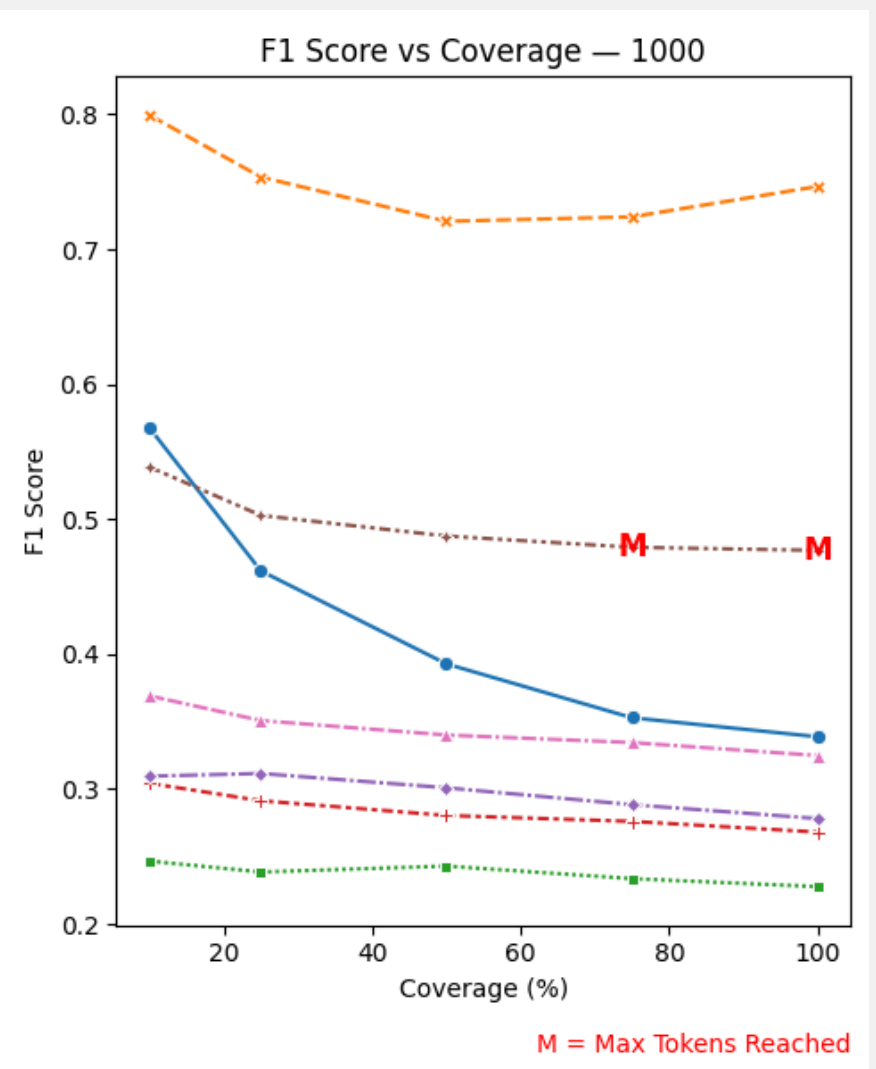
Precision



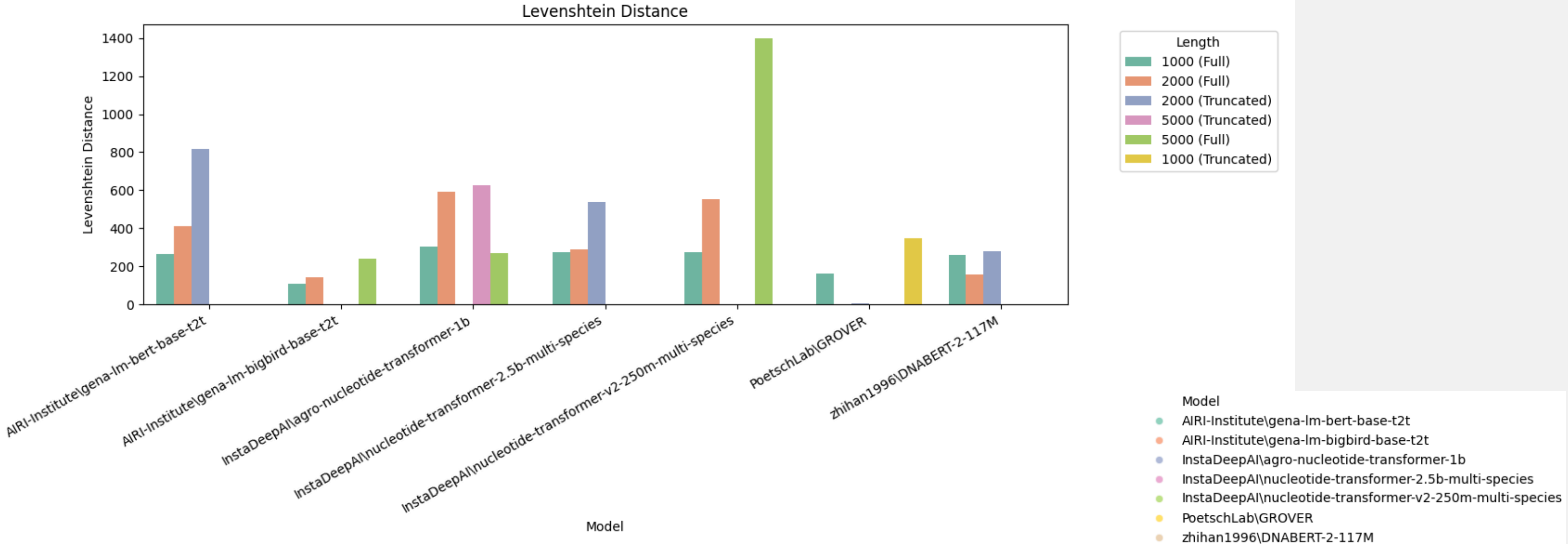
Recall



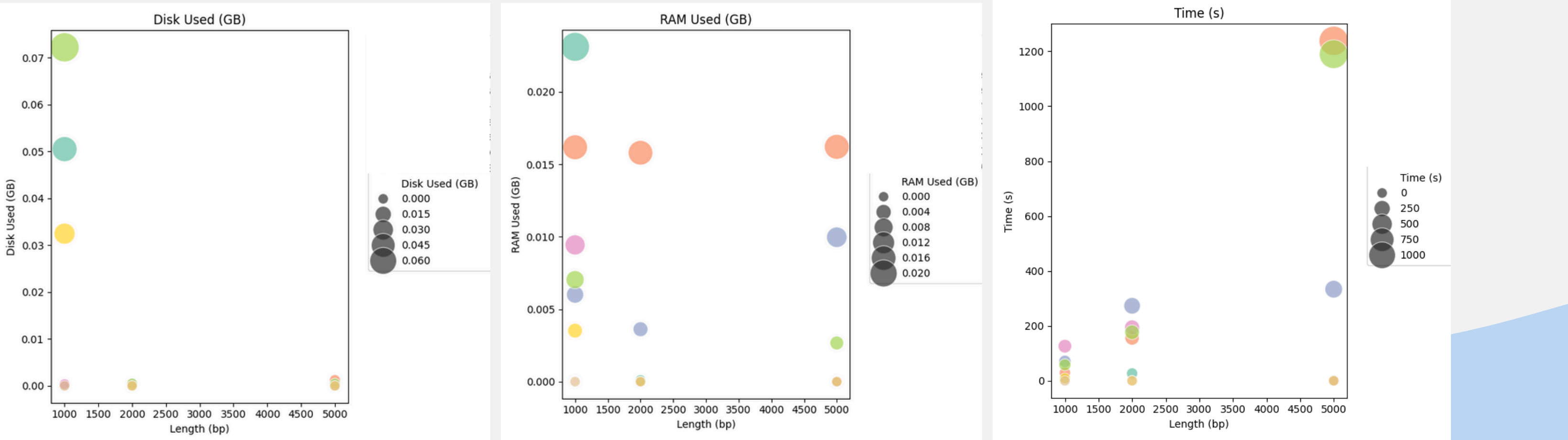
F1 Score



Levenshtein Distance



Computational Metrics



05

Integration into a Gap-Filling Pipeline - Dataset

Contigs

```
>AP012051.1_contig31
length=190651
original_pos=1298041-
1488692
TTGCATATAA...
>AP012051.1_contig32...
```

- FASTA files
- 50 - 200 contigs
- 1Kbp - 200Kbp

Simulated Draft Genomes

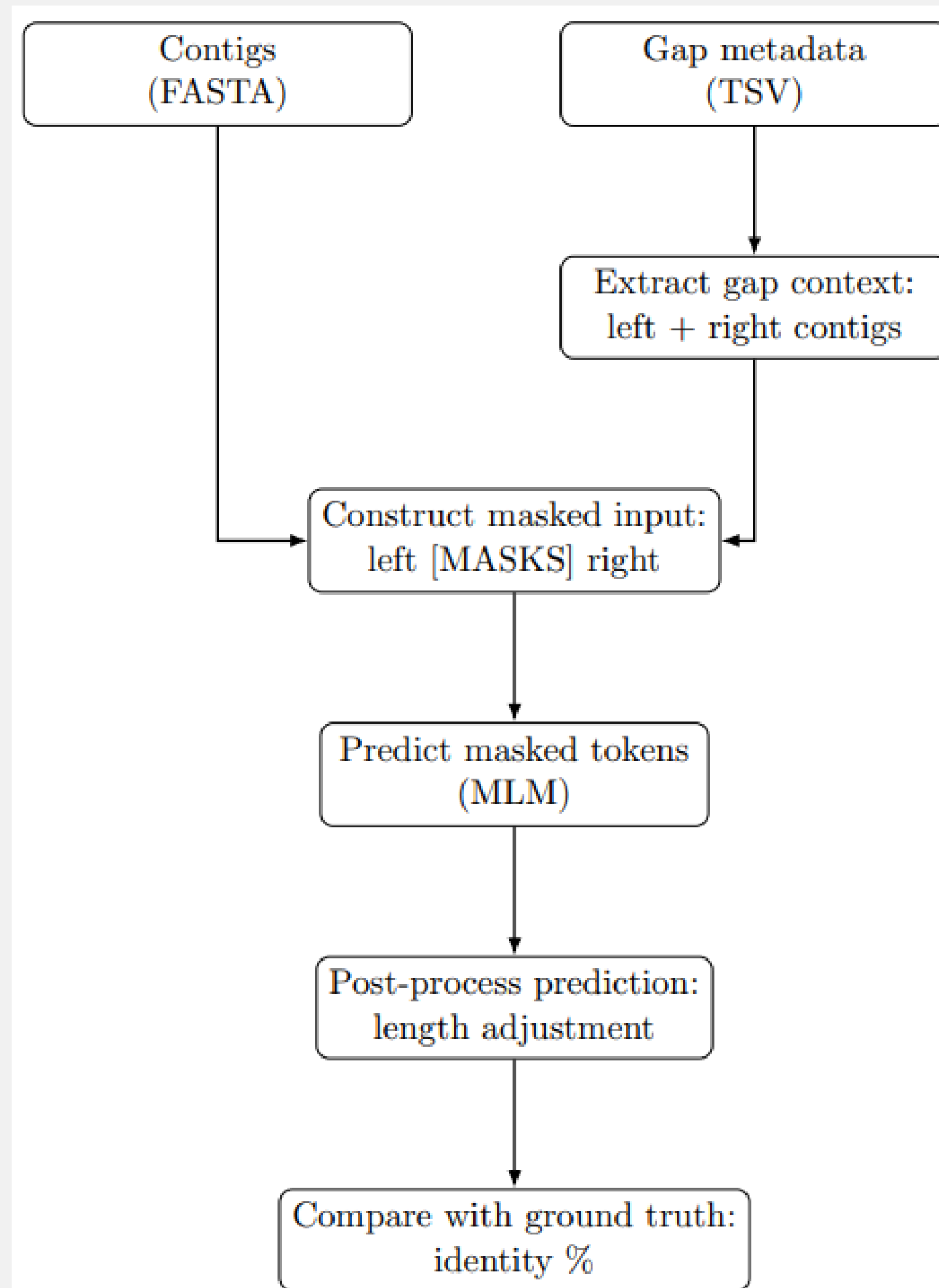
Gaps

```
gap_id start end
length sequence
AP012051.1_gap1 5951
6821 870 TTTTCAT...
AP012051.1_gap2 20817
25684 4867 TACATT...
```

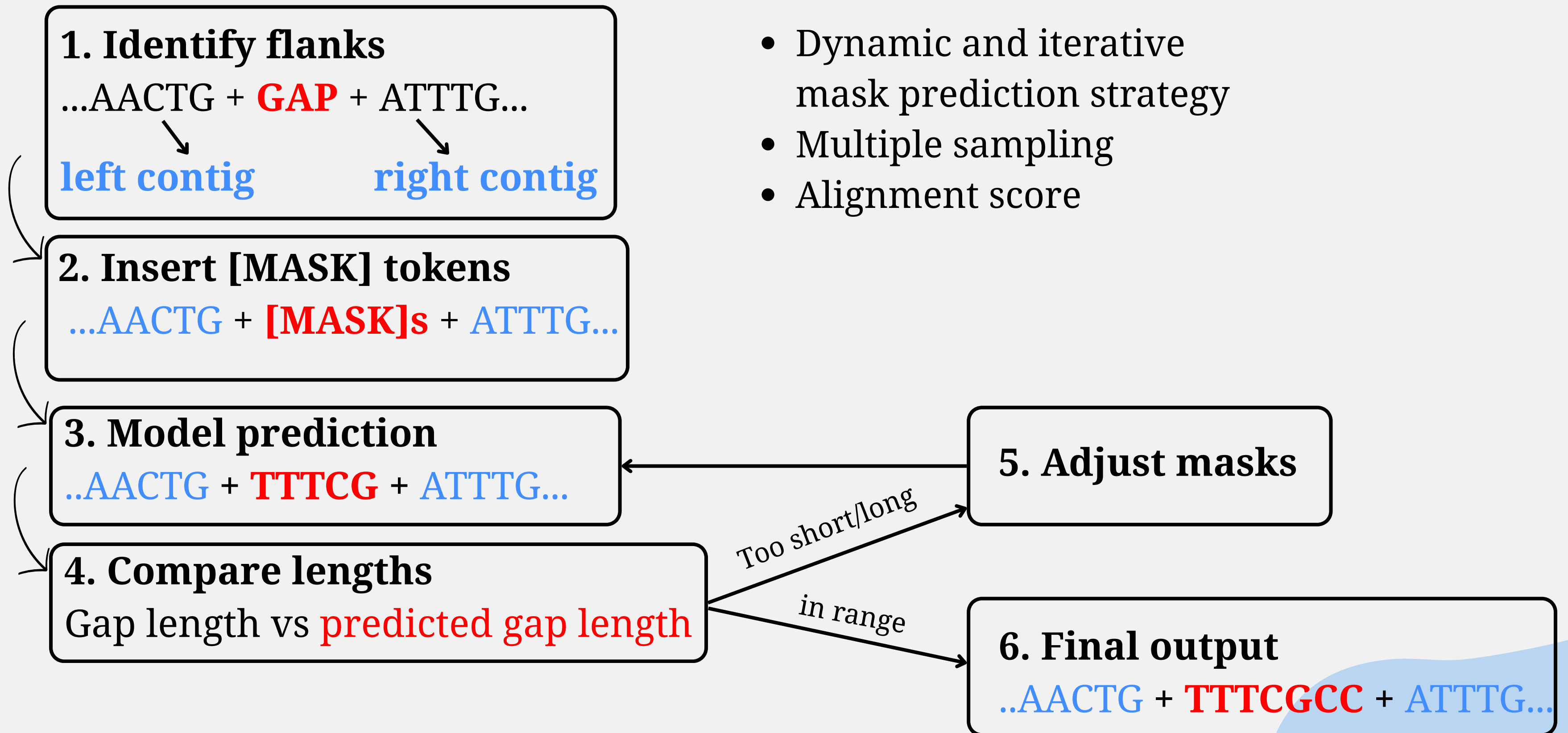
- TSV files
- 0.5Kbp - 5Kbp

05

Integration into a Gap-Filling Pipeline - Workflow



Mask Prediction Strategy



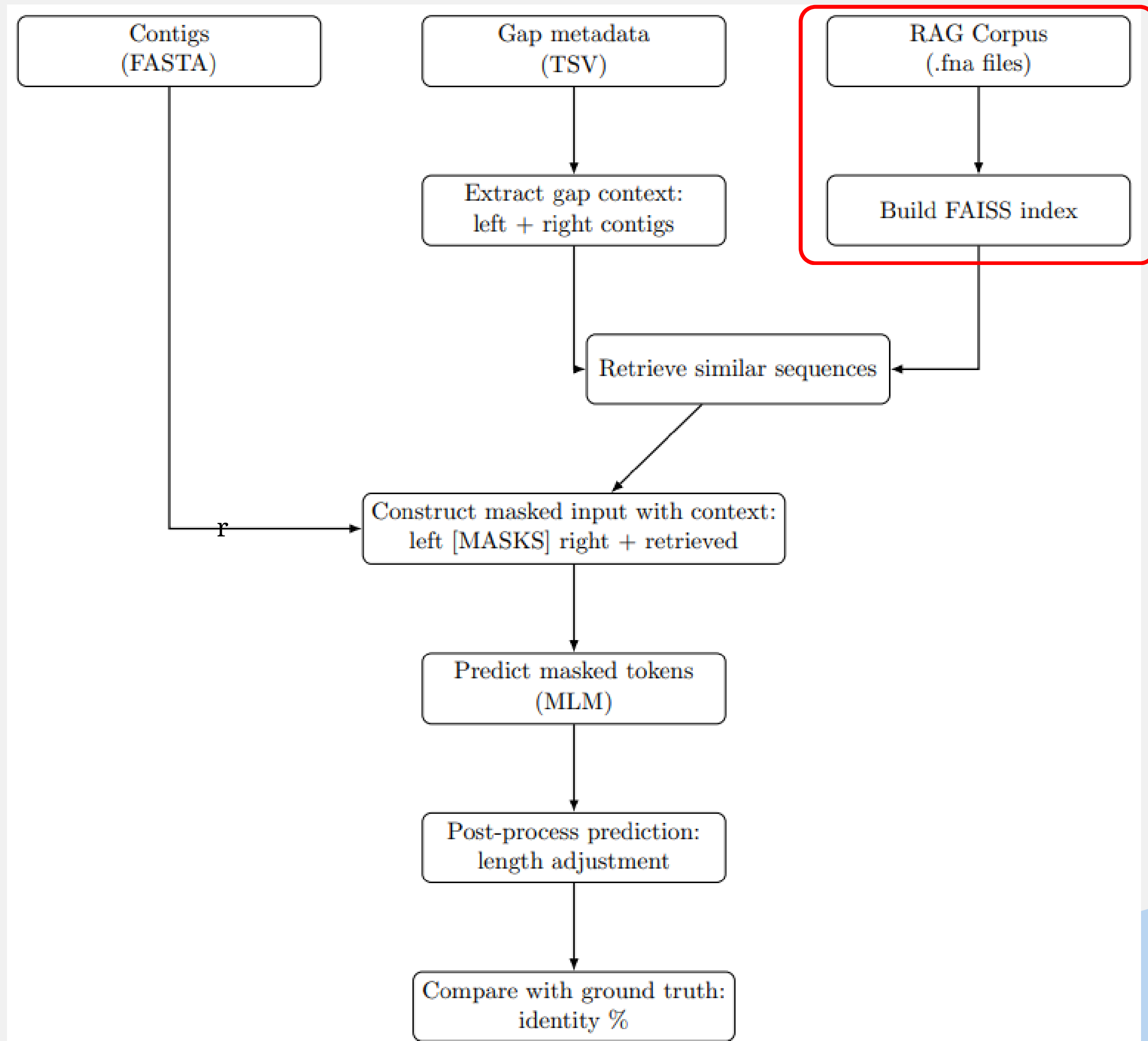
Results

Gap ID	#	Predicted	Real	Gap Length	Identity (%)						
AP012051.1_gap1	1	CTC...	TTT...	870	57.06	AP012051.1_gap2	4	CAC...	TAC...	4867	54.20
AP012051.1_gap1	2	GTC...	TTT...	870	56.95	AP012051.1_gap2	5	CAC...	TAC...	4867	54.30
AP012051.1_gap1	3	GTC...	TTT...	870	56.83	AP012051.1_gap2	6	CAC...	TAC...	4867	54.24
AP012051.1_gap1	4	GTC...	TTT...	870	57.01	AP012051.1_gap2	7	GAC...	TAC...	4867	54.18
AP012051.1_gap1	5	GTC...	TTT...	870	57.00	AP012051.1_gap2	8	CCA...	TAC...	4867	54.22
AP012051.1_gap1	6	CTC...	TTT...	870	56.67	AP012051.1_gap2	9	GAC...	TAC...	4867	54.24
AP012051.1_gap1	7	GTC...	TTT...	870	56.88	AP012051.1_gap2	10	GAC...	TAC...	4867	54.18
AP012051.1_gap1	8	GTC...	TTT...	870	57.04	AP012051.1_gap3	1	CAA...	GAG...	936	43.91
AP012051.1_gap1	9	CTC...	TTT...	870	57.11	AP012051.1_gap3	2	GAA...	GAG...	936	44.02
AP012051.1_gap1	10	GTC...	TTT...	870	56.93	AP012051.1_gap3	3	CAA...	GAG...	936	43.91
AP012051.1_gap2	1	CCA...	TAC...	4867	54.22	AP012051.1_gap3	4	CAA...	GAG...	936	43.91
AP012051.1_gap2	2	CAC...	TAC...	4867	54.22	AP012051.1_gap3	5	CAA...	GAG...	936	43.91
AP012051.1_gap2	3	GAC...	TAC...	4867	54.26	AP012051.1_gap3	6	GAA...	GAG...	936	44.02
						AP012051.1_gap3	7	GGA...	GAG...	936	44.02
						AP012051.1_gap3	8	CAA...	GAG...	936	43.91
						AP012051.1_gap3	9	CAA...	GAG...	936	43.91
						AP012051.1_gap3	10	CAA...	GAG...	936	43.91

Table 4.1: Example predictions for gaps in genome *AP012051.1*.

06 RAG Integration

- Species identification (NCBI)
- Selection of top species
- Local RAG corpus construction

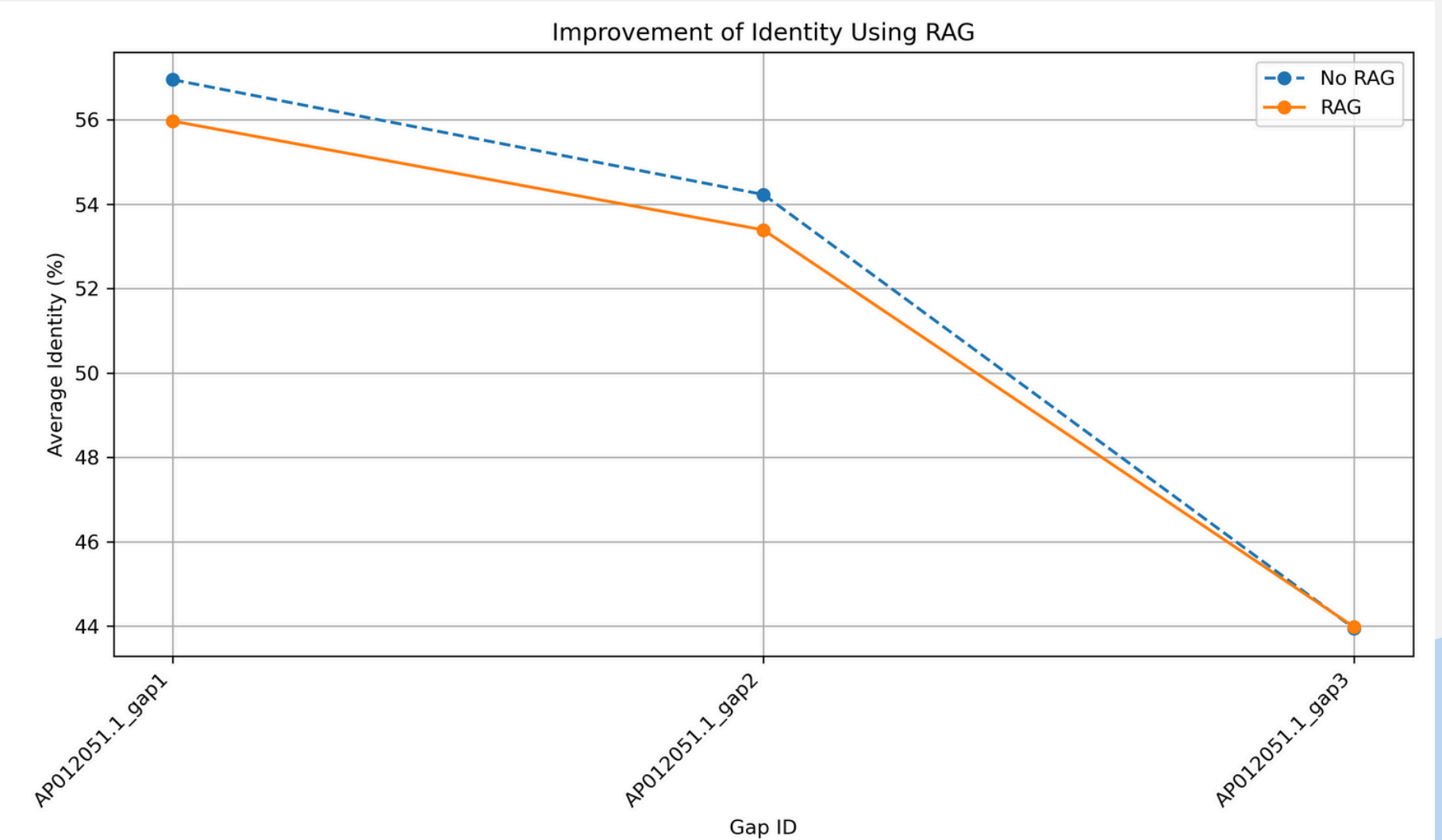


Results

Gap ID	#	Predicted	Real	Gap Length	Identity (%)
AP012051.1_gap1	1	GTC...	TTT...	870	55.98
AP012051.1_gap1	2	GGT...	TTT...	870	55.86
AP012051.1_gap1	3	GTC...	TTT...	870	56.32
AP012051.1_gap1	4	CTC...	TTT...	870	55.75
AP012051.1_gap1	5	CTC...	TTT...	870	55.75
AP012051.1_gap1	6	GTC...	TTT...	870	56.44
AP012051.1_gap1	7	GTC...	TTT...	870	55.75
AP012051.1_gap1	8	CTC...	TTT...	870	56.32
AP012051.1_gap1	9	GGT...	TTT...	870	55.75
AP012051.1_gap1	10	CTC...	TTT...	870	55.75
AP012051.1_gap2	1	CCT...	TAC...	4867	53.38
AP012051.1_gap2	2	CTC...	TAC...	4867	53.40
AP012051.1_gap2	3	GTC...	TAC...	4867	53.39
AP012051.1_gap2	4	GGT...	TAC...	4867	53.38
AP012051.1_gap2	5	GTC...	TAC...	4867	53.39
AP012051.1_gap2	6	GCT...	TAC...	4867	53.37
AP012051.1_gap2	7	CTC...	TAC...	4867	53.40
AP012051.1_gap2	8	GGT...	TAC...	4867	53.37
AP012051.1_gap2	9	GCT...	TAC...	4867	53.40
AP012051.1_gap2	10	GTC...	TAC...	4867	53.40

AP012051.1_gap3	1	GAA...	GAG...	936	44.02
AP012051.1_gap3	2	GAA...	GAG...	936	44.02
AP012051.1_gap3	3	GAA...	GAG...	936	44.02
AP012051.1_gap3	4	CAA...	GAG...	936	43.91
AP012051.1_gap3	5	GGA...	GAG...	936	44.02
AP012051.1_gap3	6	CAA...	GAG...	936	43.91
AP012051.1_gap3	7	GAA...	GAG...	936	44.02
AP012051.1_gap3	8	GGA...	GAG...	936	44.02
AP012051.1_gap3	9	CAA...	GAG...	936	43.91
AP012051.1_gap3	10	CAA...	GAG...	936	43.91

Table 4.4: Example predictions for gaps in genome *AP012051.1* using RAG.



07 Agent-based Gap-Filling Pipeline

Cloud

Gemini-2.5-flash

```
model = init_chat_model("gemini-2.5-flash", model_provider="google_genai")
```

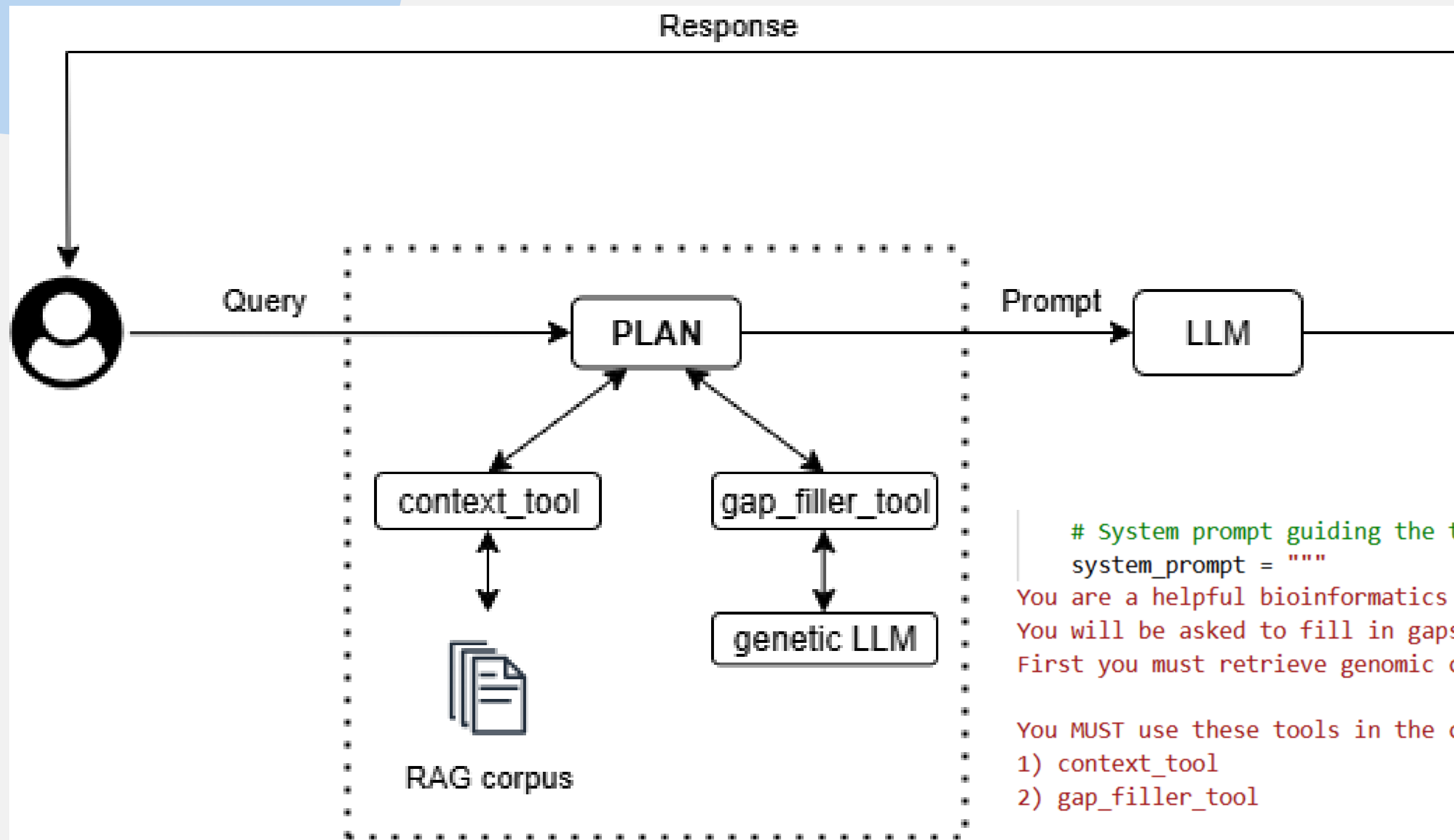
- Accessed via Google API key
- Supports tool calling natively
- Integrated with LangChain
- ☒ Fast, scalable, advanced model
- ☒ Flexible pipeline orchestration

Local

LLama.cpp - Llama 2 7B

```
docker run -v "C:\Users\clari\...\models" \ -p 8000:8000  
ghcr.io/ggml-org/llama.cpp:server \ -m /models/llama-2-7b-  
model.gguf --port 8000 --host 0.0.0.0 -n 512
```

- Deployed via Docker (official GGML image)
- Model: llama-2-7b-chat.Q5_K_M.gguf
- Exposed through REST API + web UI
- Integrated with LangChain
- ☒ Privacy & full control
- ☒ Easy integration with custom tools



- **Agent:** planner
- **Tools:** context_tool and gap_filler_tool

```
# System prompt guiding the tool order and usage
```

```
system_prompt = """
```

```
You are a helpful bioinformatics agent that solves DNA sequence problems using tools.  
You will be asked to fill in gaps in genomic sequences, where gaps are marked with "---"  
First you must retrieve genomic context, then you must fill the gaps.
```

```
You MUST use these tools in the correct order:
```

- ```
1) context_tool
2) gap_filler_tool
```

```
CRITICAL RULES:
```

- ```
- The user provides an integer `gap_length` that represents the intended gap size.  
- You MUST pass EXACTLY that `gap_length` to gap_filler_tool (do not recompute it).  
- Use context_tool first to retrieve up to three best matching contexts.  
- Then call gap_filler_tool with:  
    sequence=<the sequence containing '---'>,  
    rag_matches=<the contexts you just retrieved>,  
    gap_length=<the exact integer provided by the user>.  
- Be concise in your final answer.
```

```
"""
```

08 Future Work

Explore AR Models

- Explore and evaluate AR models
- Create a gap prediction pipeline for AR models

Improve RAG

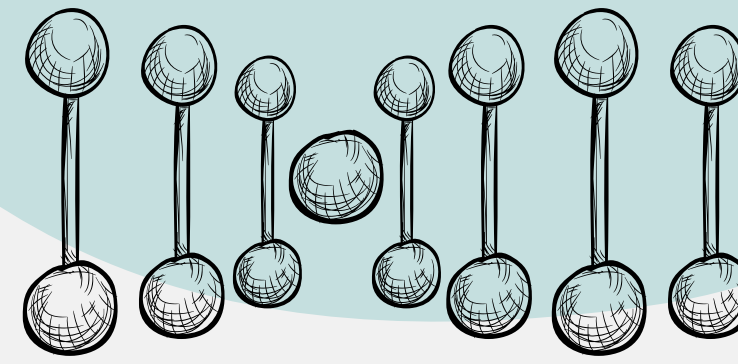
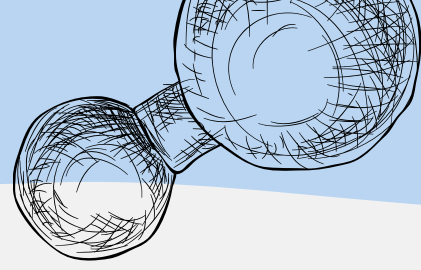
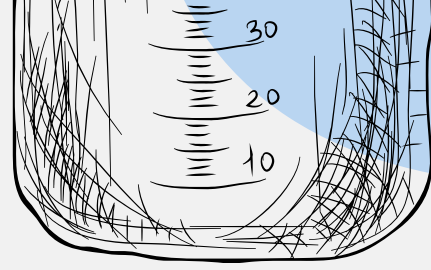
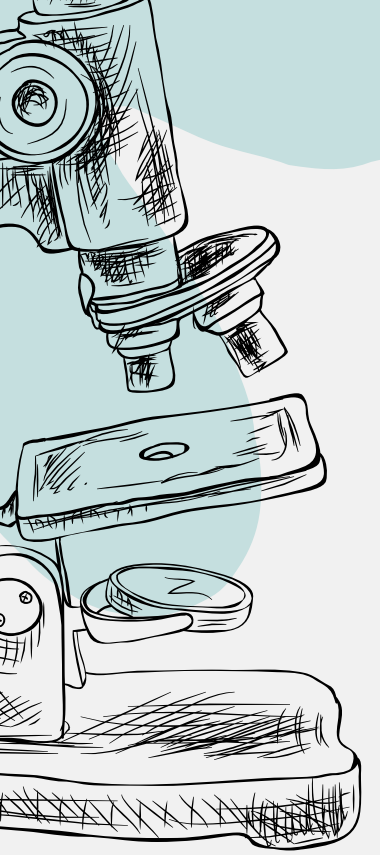
- Expand corpus
- Use FAISS in agent-based pipeline
- Remove dependency on gap length

Scalability

- Leverage more powerful GPUs
- Use more powerful LLM models
- Explore cloud deployment for storage

Improve End-to-end System

- Integrate more agents
- Integrate MCP protocol
- Provide REST API to facilitate integration into lab workflows



Thank you!

